PREFACE

Why are radio communications important to understand as an aspiring hacker guy?

When initially pondering this question, really the subject of radio communications in general, it is easy to think of reconnaissance. Information gathering, reconnaissance, surveillance, all of these things are certainly important to an attacker. After all, if radio waves are everywhere in the air around us, why not just receive that data and see for ourselves what it entails?

However, the manipulation of ANY mechanism of data transmission is crucial to exploitation. Whether the medium is over the air or via CAT5 cables, the concept is the same. The essence lies in comprehending the data's intended purpose and its role within a system. By understanding what the data is meant to achieve, attackers can identify opportunities to leverage this process for unauthorized access. This principle applies universally, underscoring the critical importance of securing data in transit.

For instance, the advent of the Internet of Things (IoT) and the proliferation of wireless communication in controlling critical infrastructure have opened new avenues for exploitation. Critical infrastructure is oftentimes managed remotely via wireless technology.

The main point being...

The principles that underpin all wireless communication are fundamentally rooted in the basic concepts of radio communication, which involve the transmission and reception of electromagnetic waves through the air.

Understanding this process is crucial to developing a foundation of understanding data transmission prior to computing in order to effectively understand how modern systems transmit data. Then, it is possible to begin understanding how to exploit said systems.

HF: Around the Corner to Around the World

Before SATCOM technology existed, HF radios were the only means of communicating to ships at sea. The fact that HF can communicate beyond the horizon makes it an indispensable tool for long distance ship-to-ship and ship-to-shore messages. Likewise, before the days of the transatlantic cables, HF (or short wave) radios were the only way to talk between continents. Today they are still used to share the overall burdens of long distance communications. But the unique virtue of HF radio has also created some challenges. Worldwide radio transmissions are easy to intercept and the HF spectrum is complicated by signals emanating from the many individual transmitters located around the world. Special techniques must be implemented in the radios to take advantage of the radio's long range, while still preserving the clarity of the channel and reduce interception.

Encryption reduces unfriendly utilization of intercepted signals and sophisticated coding schemes helps fight through clutter, but these techniques can reduce throughput (compared to that of a clear channel). Nevertheless, HF radios still play an indispensable role in the communicator's tool kit. HF manpack radios with various antenna options can cover a practical range of from "around the corner" to "around the world." Although some long range communications are now transmitted via satellite, HF still has the advantage of not requiring (or relying on) any infrastructure.

High Frequency radio communication operates in the frequency range of 3 to 30 MHz. This band of frequencies is particularly useful for long-distance communication because HF signals can be reflected off the ionosphere, a layer of electrically charged particles in the Earth's atmosphere. This phenomenon, known as "skywave" or "skip" propagation, allows HF radio waves to travel beyond the horizon, covering distances of several thousand kilometers.

It is especially valuable in remote areas where other forms of communication (like telephone or internet) might be unavailable or unreliable. HF radio is known for its ability to maintain communication in disaster situations or in environments where other systems might fail, making it an essential tool for emergency communication.

Satellites have deemphasized the use of HF radio and as a result many present day communicators do not have an understanding of what modern HF capabilities are

Line of Sight: Typically less than 30km range, is limited by terrain obstructions and curvature of the earth

Range = function of operating frequency, power level, and antenna height

Restrict range → reduce area interference and ease frequency reuse requirements

Surface or Ground Wave:

50 km on land, 300+ km over the sea

Range depends on operating frequency and terrain obstructions

Requires vertically polarized antennas because…

Reduced Ground Losses: Vertically polarized waves tend to suffer less attenuation from ground losses compared to horizontally polarized waves when traveling over land and sea. This is because the electric field in a vertically polarized wave is perpendicular to the Earth's surface, reducing the energy absorbed by the ground.

Consistent Signal Strength: Vertically polarized signals maintain relatively consistent signal strength over various terrains, including over water, urban, and rural landscapes, making them reliable for ground-based communication.

Less Susceptible to Fading: Vertically polarized signals are generally less susceptible to multipath fading caused by reflections off the Earth's surface or other objects, which can be a significant issue for horizontally polarized waves.

Compatibility with Mobile and Marine Applications: Many mobile and marine communication systems use vertically polarized antennas because they are more suited to the changing orientation of vehicles and vessels. This ensures more stable communication regardless of the vehicle's or vessel's pitch and roll.

Beyond Line of Sight: range about 400 km using Near Vertical Incidence skywave (NVIS)

The key feature of NVIS is the use of high-angle radiation, where radio waves are transmitted almost vertically towards the ionosphere. Upon reaching the ionosphere, these waves are refracted back to Earth, covering an area in the transmitter's immediate vicinity that would otherwise be in a communication dead zone due to the curvature of the Earth or obstructions like mountains and buildings.

BLOS can be used when satellites are down

- Terrain obstructions not a factor
- Horizontally Polarized Antennas

- Frequency range < 10 MHz
- Voice or low rate data, data rate depends on waveform
- Operating frequency is determined by ionospheric conditions and solar cycles
- Automatic link establishment helps solve the operating frequency problem

Automatic Link Establishment (ALE) enables radio stations to connect with each other automatically, choosing the optimal frequency for communication without requiring manual intervention. ALE is primarily used in the HF (High Frequency) bands.
Here's how ALE works and its key features:

ALE systems scan a predefined set of frequencies to determine the best available channel for communication. This process involves both listening for signals from other stations and transmitting calls on various frequencies.

ALE adapts to changing propagation conditions by continuously monitoring the frequencies and the quality of the links. If the current link deteriorates, ALE can automatically switch to a better frequency, ensuring stable communication.

ALE systems perform "soundings," which are essentially calls sent out on various frequencies to assess which are most suitable for communication at any given time. Other ALE-equipped stations listen to these soundings and respond, allowing each station to build a database of viable frequencies for each potential contact.

Digital Select Calling (DSC): ALE uses a form of digital select calling to establish a link, where each station has a unique address. When a station wants to establish a link, it sends out a call that includes the address of the target station. Only the station with the matching address responds, reducing the chance of interference and unauthorized access.

Long Range Communications: Over 4000 km

Range depends on antenna power level and atmospheric conditions

ALE is critical for this range of communication

Note: HF is also known as short wave

Radio Technology Beginnings = Clerk Maxwells Treatise on Electricity and Magnetism in 1873

In 1888 Heinrich Hertz after whom the measurement of frequency is named, performed experiments that showed disturbances caused by a spark coil showed characteristics of Maxwells radio waves. Guglielmo Marconi performed experiments with wireless telegraphy using morse code and by 1896 Marconi had communicated messages over a few kilometers' distance.

At that time, it was believed that radio waves in the atmosphere travelled in straight lines and that they would not be useful in Over-The-Horizon (OTH) communication

In 1901 Marconi disproved this by using an antenna made of 120m of wire attached to a simple kite in the air, where he detected a telegraphic signal transmitted from Cornwall, England (detected in Newfoundland, Canada)

Edward Appleton then discovered the ionosphere, a blanked of electrically charged ionized particles capable of reflecting radio waves

By World War 2, HF radio was the primary means of long haul communication
Simplified

1873 Clerk Maxwell Radio Wave Theory → 1888 Heinrich Hertz generates radio waves with coils → Marconi experiments wireless telegraphy, successful over a few kilometers in 1896 and 3000km in 1901 → Edward Appleton discovers the ionosphere and its ability to reflect radio waves

Satellites caused a temporary hiatus of usage of HF radio

Fear of a single point of failure led to revamped interest in HF

Electromagnetic radiation

Radio waves = electromagnetic spectrum

Electromagnetic family includes x ray, ultraviolet, and visible light

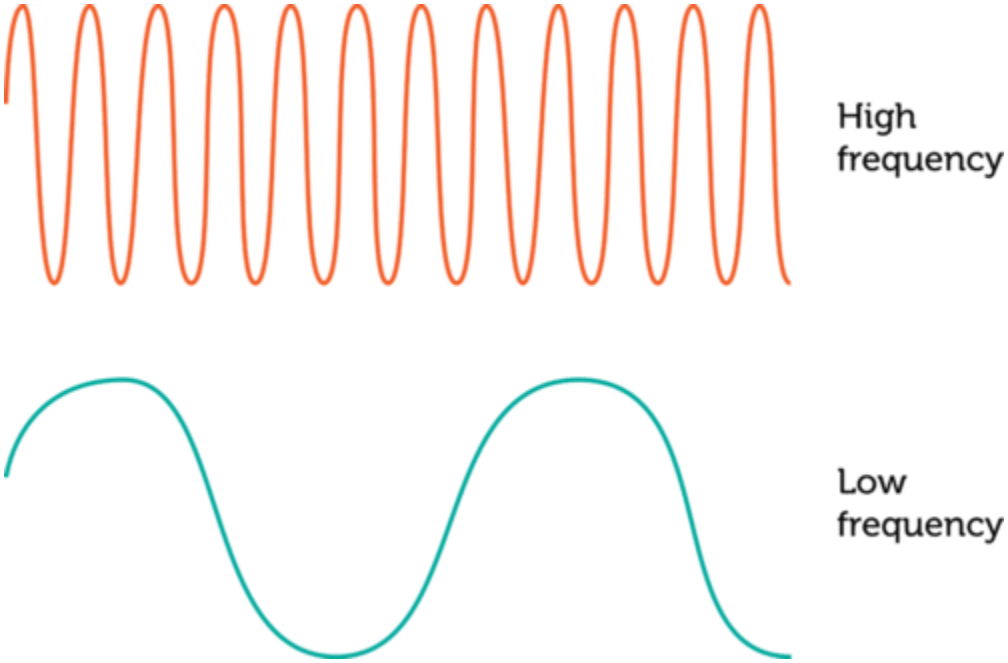
Radio waves propagate at the speed of light

Radio wave characterization = 3 factors

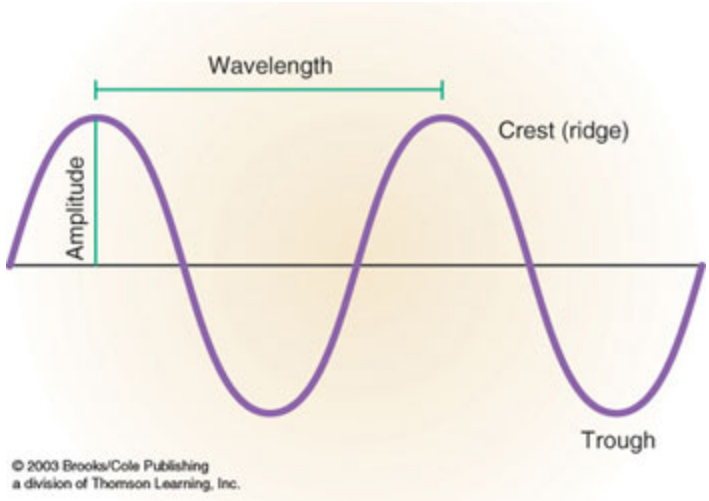
Amplitude, Frequency and Wavelength

Amplitude = strength, distance between peak and lowest point and is measured in volts, usually expressed in terms of an average value called root-mean-square

Frequency = number of repetitions or cycles it completes in a given time, measured in Hertz where a Hertz equals the number of cycles per second a wave occurs



Wavelength is the distance between crests of a wave measured in meters



Speed of Propagation = wavelength x frequency

Mathematically represented as: $c=f\times\lambda$

Where c is a constant equal to the speed of light (3.00 x 10^8 power meters per second in a vacuum)

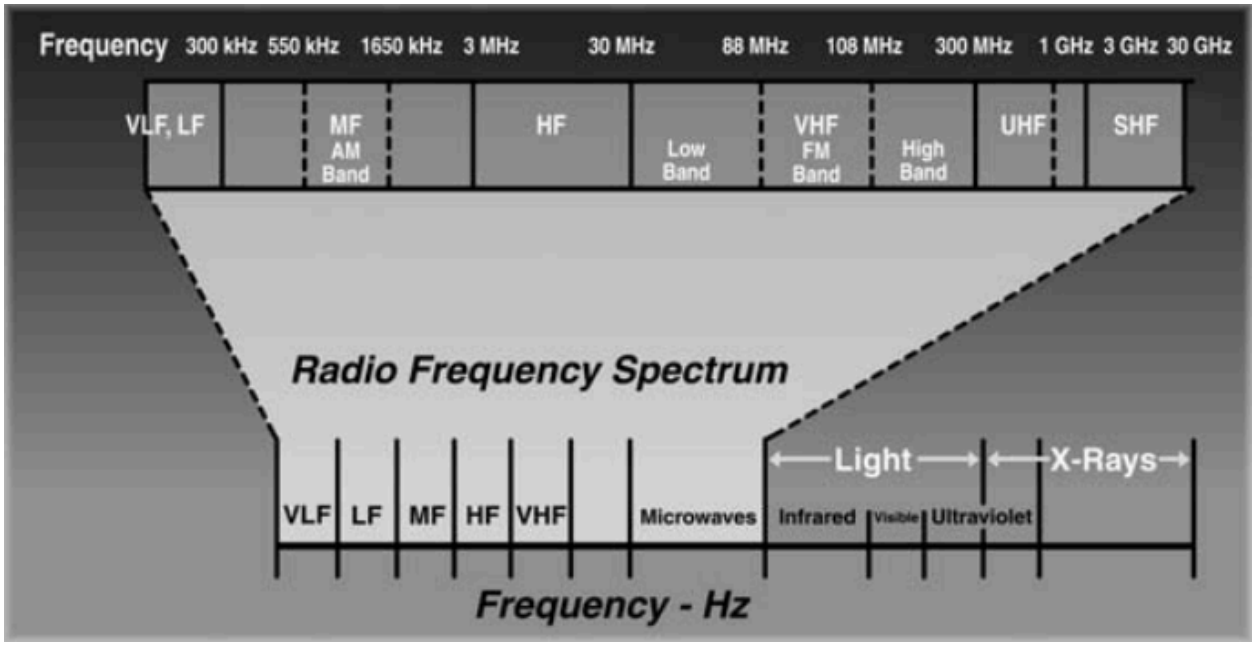
f is frequency of a wave

The lambda (λ) is the symbolic representation of wavelength

It should be noted that air IS NOT a vacuum and so the speed of radio waves is actually slightly less

Because the speed of light (speed of propagation) is a constant unchanging variable, if either frequency or wavelength increases, the other must decrease. This negative relationship between frequency and wavelength is critical to understand.

In more intuitive terms, if you consider that the wave must travel at a fixed speed, then fitting more cycles (higher frequency) into a given period requires each cycle to be shorter (shorter wavelength). Conversely, if the cycles are longer (longer wavelength), there can only be fewer of them in the same period (lower frequency). This is why as the wavelength increases, the frequency decreases, and this relationship is a cornerstone of wave behavior across various physical phenomena.



Usable frequency for radio is from 20kHz to 30 GHz or 30,000 MHz

To go up, divide by 1,000, to go down multiply by 1,000

For instance if I wanted to convert 30 GHz to MHz, billions to millions, I multiply 30 by 1000 to get 30,000 MHz and vice versa

Terahertz THz (trillion), Petahertz PHz (quadrillion) and Exahertz EHz (quintillion)also exist, but pertain more to infrared, visible light, UV, x rays and gamma rays

HF is defined as the frequency range between 3 to 30 MHz, most radios HF use 1.6 to 30 MHz

Long haul HF comms → 4 to 18 MHz

The reason for the alias short term for HF comes from their “short” wavelength of 10 to 100 meters were less than radio broadcast stations in early days

Interesting Transmission Type
Facsimile or FAX: In traditional telephone line fax machines, the frequency modulated audio tones are transmitted within the voice frequency band of the telephone network, typically between 300 Hz and 3,400 Hz. For internet fax services, the data is transmitted over digital networks without a direct relationship to specific frequency bands.

Frequency Allocation and Modulation
By themselves radio waves are simply a rhythmic stream of continuous waves containing no data

Radio waves are modulated to carry data by carrier waves, modulation refers to changing the amplitude of a wave

The carrier wave is a continuous electromagnetic wave of a constant amplitude and frequency

Phase = measurement of a complete wave cycle

Simplest method of modulating a carrier is turning the signal off and on by means of a telegraph key, known as on-off keying (OOK)

Carrier Wave: Imagine a continuous wave being transmitted at a constant frequency and amplitude; this is your carrier wave. In the absence of any modulation, this wave would just continuously propagate without conveying any useful information.

Telegraph Key: The telegraph key is a simple switch that connects and disconnects the circuit transmitting the carrier wave. When the key is pressed down, the circuit is closed, and the carrier wave is transmitted (the light is "on"). When the key is released, the circuit opens, and the transmission of the carrier wave stops (the light is "off").

Transmission of Information: Information, such as Morse code, is encoded in the pattern of "on" and "off" states. For Morse code, a short press of the key sends a "dot," and a longer press sends a "dash." The sequence of dots and dashes, which correspond to letters and numbers, is sent over the airwaves by the presence or absence of the carrier wave.

Reception: The receiving end detects the presence or absence of the carrier wave and decodes the pattern back into the original message, such as a Morse code transmission being translated back into text.

More Modulation Methods:

Amplitude Modulation (AM), Single Side Band (SSB), and Frequency Modulation (FM) are all methods of modulating a carrier wave to transmit information via radio waves. Each technique varies specific characteristics of the carrier wave in different ways to encode the message.

Amplitude Modulation (AM)
In AM, the amplitude (strength) of the carrier wave is varied in proportion to the amplitude of the message signal (the information you want to transmit), while the frequency and phase of the carrier remain constant.

The result is a modulated signal that consists of the carrier frequency plus upper and lower sidebands. The sidebands contain the actual information and are symmetrically spaced above and below the carrier frequency, each mirroring the other.

AM is susceptible to noise and interference because many sources of electrical noise also vary in amplitude and can therefore add unwanted signals to the AM signal.

Single Side Band (SSB)
SSB is a refinement of AM that eliminates one of the sidebands and sometimes the carrier as well. This makes SSB more bandwidth-efficient and power-efficient than standard AM.
There are two types of SSB: Upper Sideband (USB) and Lower Sideband (LSB). USB transmits only the upper sideband, while LSB transmits only the lower sideband.
SSB is used in long-distance communication, such as amateur (ham) radio and aviation, because it requires less bandwidth and transmits power more efficiently than conventional AM.

AM Equivalent (AME or sometimes denoted as DSB-FC)
AM Equivalent, or Double Sideband Full Carrier (DSB-FC), refers to the standard form of AM where both the upper and lower sidebands are present, along with the carrier. It's *the most basic form of AM and is what most people refer to when they talk about AM*.

This form of AM is not as efficient as SSB because it uses more bandwidth (due to both sidebands being transmitted) and more power (to transmit the carrier and both sidebands).

Frequency Modulation (FM)

In FM, the frequency of the carrier wave is varied in accordance with the amplitude of the message signal, while the amplitude of the carrier wave remains constant. The greater the amplitude of the input signal, the greater the frequency deviation of the carrier wave from its baseline.

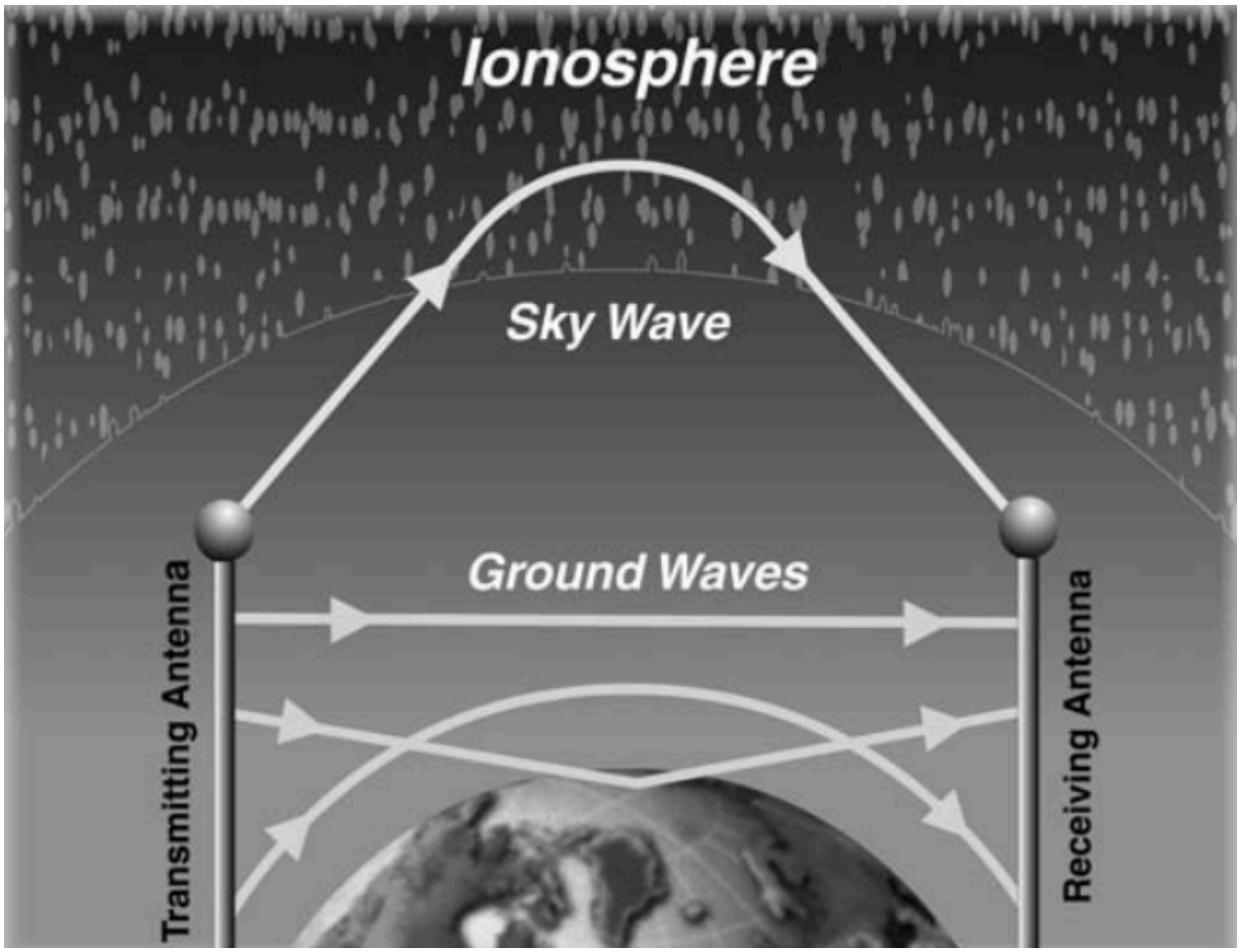
FM is less susceptible to noise and interference than AM because variations in amplitude (which is where noise typically manifests) do not affect the frequency of the carrier wave.

FM is widely used for high-fidelity broadcasts such as FM radio broadcasting, where audio quality is important, as well as in two-way radio communication systems for its resistance to electrical noise.

Overall,

FM = better and cleaner signal but requires more bandwidth

Narrowband FM = less bandwidth and less signal quality


Ionosphere
Sky Wave
Ground Waves
Transmitting Antenna
Receiving Antenna

Propagation = how radio signals radiate outward from a transmitting source

Radio waves do not often travel in straight lines

Ground waves have 3 components

1. Surface waves
2. Direct waves
3. Ground reflected waves

Ground waves, which are radio waves that propagate along the Earth's surface, are composed of three main components: surface waves, direct waves, and earth-reflected waves. Each of these components plays a role in how radio signals travel over the ground, affecting their range and clarity.

Surface Waves: These are the radio waves that travel along the surface of the Earth, following its contours. Surface waves are particularly significant at lower frequencies, where they can efficiently travel over considerable distances by continually bending around the Earth's curvature due to diffraction. Diffraction is a phenomenon that occurs when a wave encounters an obstacle or a slit that is comparable in size to its wavelength. Imagine water waves hitting a small opening between two barriers: some of the waves will bend around the edges of the barriers and spread out as they pass through the opening. This spreading out or bending of waves around obstacles is what we call diffraction.

They are most effective for frequencies in the range of 50 kHz to 250 kHz and are commonly used for local radio broadcasting, radar, amateur radio, and radio navigation. The nature of the ground over which these waves travel, such as its conductivity and topography, significantly affects their propagation.

Direct Waves/Line of Sight: Direct waves travel in a straight line from the transmitting antenna to the receiving antenna. These are the waves that would be received if there were a clear, unobstructed line of sight between the transmitter and receiver. Direct waves become weaker as the distance increases and are a component of ground waves when the transmitting and receiving antennas are within each other's line of sight.

Earth-Reflected Waves: These are the radio waves that strike the Earth's surface and are reflected back towards the receiving antenna. The strength of the reflection and, consequently, the effectiveness of the earth-reflected waves depend on local conditions such as the ground's conductivity and the angle of incidence.

SUMMARY SO FAR:

Radio signals propagate from a transmitting antenna as waves through space at the speed of light.

Radio frequency is expressed in terms of hertz (cycles per second), kilohertz (thousands of Hertz), or megahertz (millions of Hertz).

Frequency determines the length of a radio wave; lower frequencies have longer wavelengths and higher frequencies have shorter wavelengths.

Long-range radio communications take place in the high-frequency (HF) range of 1.6 to 30 MHz. Different portions of this band are allocated to specific radio services under international agreement.

Modulation is the process whereby the phase, amplitude, or frequency of a carrier signal is modified to convey intelligence.

HF radio waves can propagate as sky waves, which are refracted from the earth's ionosphere, permitting communications over long distances.

The Effects of the Ionosphere and Solar Activity on HF Radio Propagation

The ionosphere is a region of electrically charged particles or gases in the earth's atmosphere, extending from approximately 50 to 600 km above the earth's surface.

The ionosphere's impact on HF radio propagation is significant due to its ionized layers, which can reflect or refract HF waves, facilitating long-range communication. When radio waves strike these ionized layers, depending on frequency, some are completely absorbed, others are refracted so that they return to the earth, and still others pass through the ionosphere into outer space. Absorption tends to be greater at lower frequencies, and increases as the degree of ionization increases. Solar activity, including solar flares and sunspots, dynamically alters the ionosphere's density and composition, affecting HF wave propagation. Increased solar activity can enhance ionospheric reflectivity, improving HF signal range, but can also cause signal degradation or loss due to ionospheric storms.

Ionization is the process in which electrons are stripped from atoms and produces electrically charged particles. It is a direct result from solar radiation.

The angle at which sky waves enter the ionosphere → the incident angle

Radio waves reflect at the same angle they hit the ionosphere

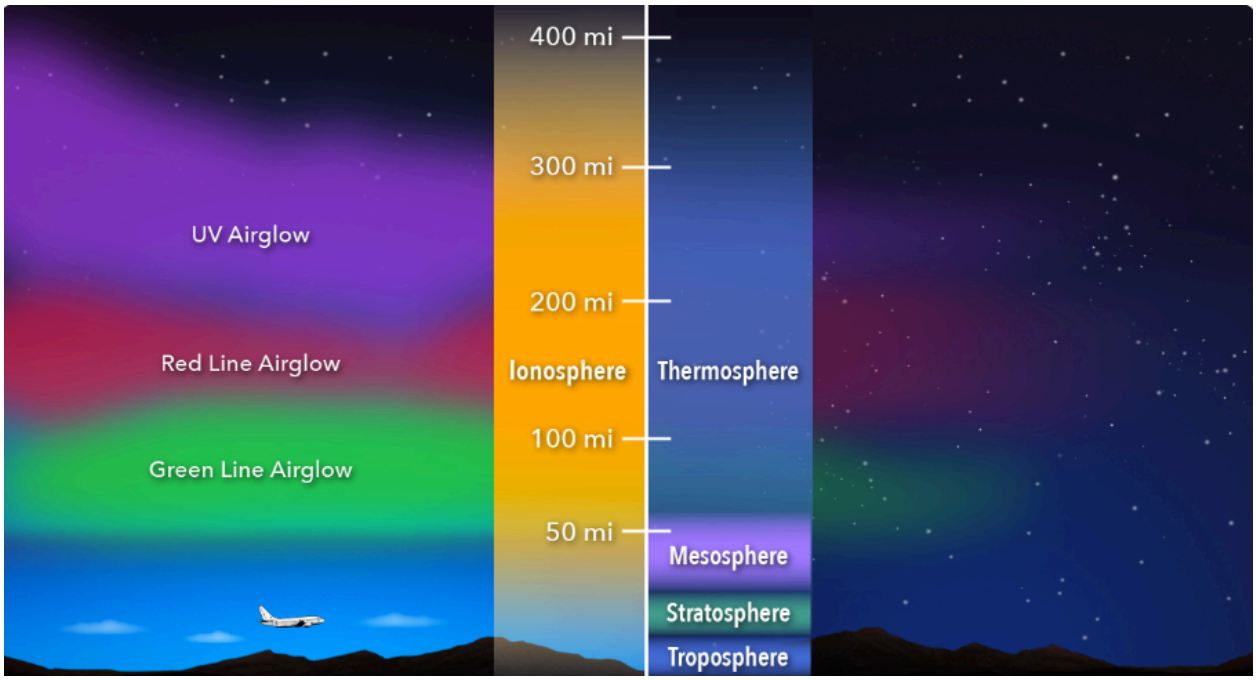
The incident angle is an important factor in determining communications range. If you need to reach a station that is relatively far from you, you would want the incident angle to be relatively large.
To communicate with a nearby station, the incident angle should be relatively small.

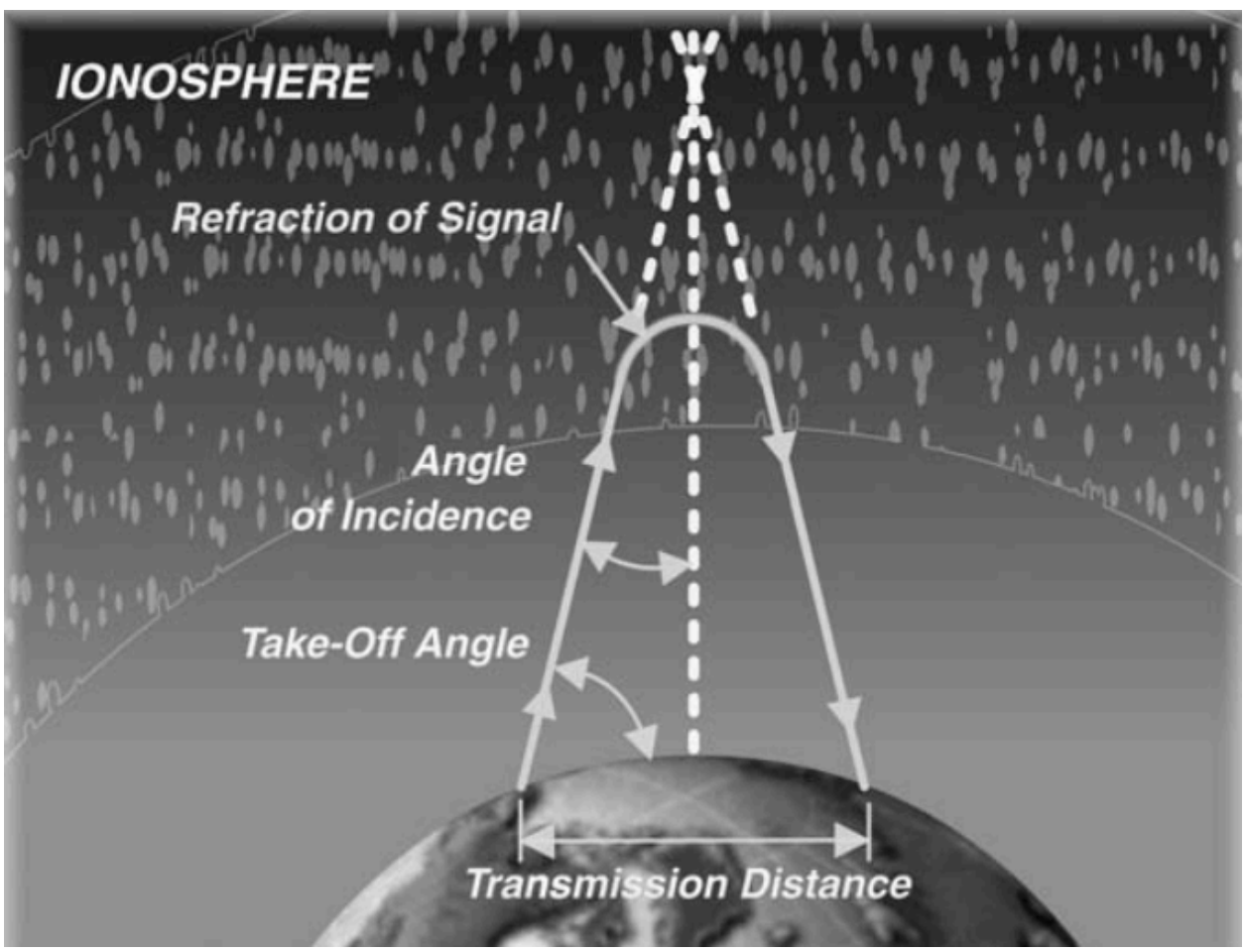
NOTE: The gases in the ionosphere may glow and be visible in a phenomena commonly known as the northern, southern lights.

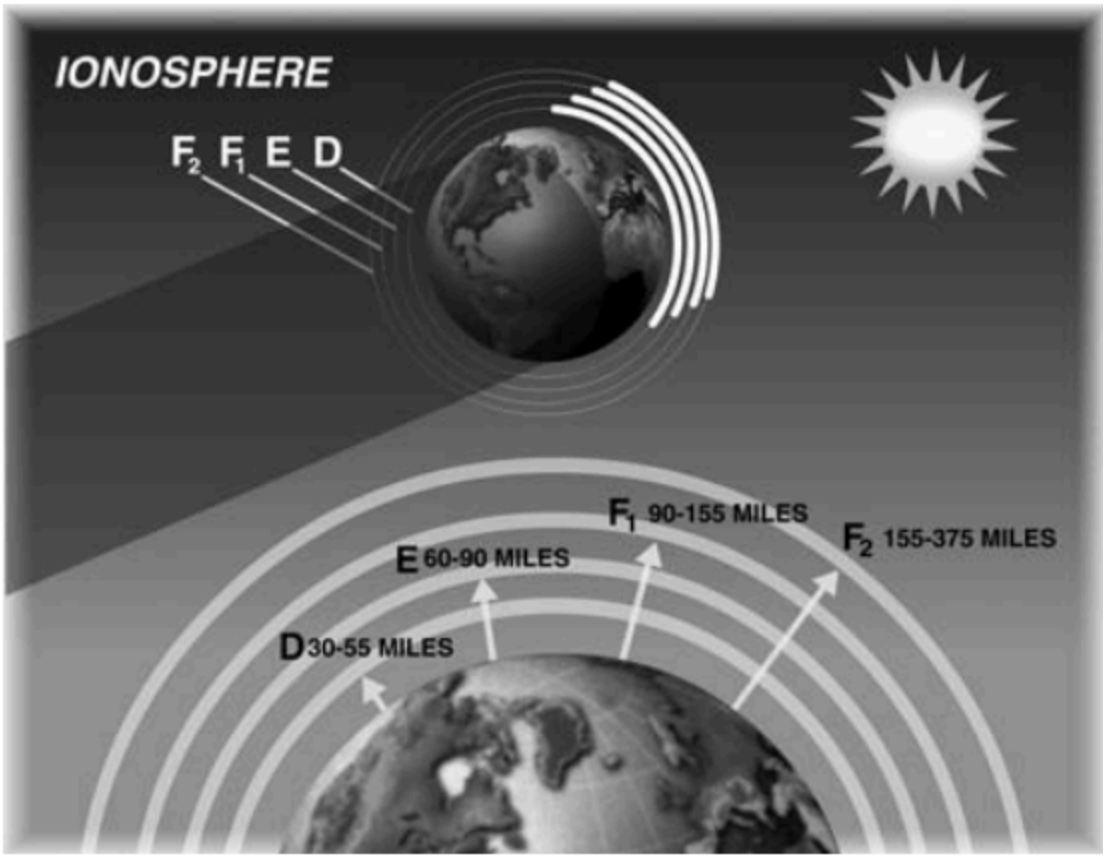
For context, the ionosphere is an outer or upper layer of the earths atmosphere.

In order of closest to farthest from earths surface,
Troposphere, Stratosphere, and Mesosphere are from the earths surface to roughly 50 miles above where the Thermosphere begins up to around 500 km where the exosphere begins

The thermosphere is where most of the ionosphere is located







E layer was what was discovered by Appleton in the early 1920s

Outer layers = more ionization, closer layers have less

More ionization = more range / efficiency

the D layer reaches maximum ionization when the sun is at its zenith

The E layer reaches maximum ionization at noon

The most heavily ionized region of the ionosphere, and therefore the most important for long-haul communications, is the F layer which retains its ionized properties even after sunset

Shorter wavelength enables penetration of the initial layers, too long of a wavelength may result in radio waves exiting the ionosphere and heading to space (same with too small of angle of incidence IE vertical transmission directly into the ionosphere)

Ionization is higher in spring and summer because of the longer days

More sunspots = more ionization

Sudden Ionospheric Disturbances are caused by solar flares and can have impacts on HF comms

Solar flares negatively impact HF comms, as they tend to neutralize its reflective properties

Maximum Usable Frequency (MUF): The MUF is the highest frequency that can be used for communication between two specific locations via ionospheric reflection at a given time. It represents the upper limit of usable frequencies for a particular path under prevailing ionospheric conditions. Frequencies higher than the MUF are typically absorbed or refracted by the ionosphere rather than being reflected back to Earth, resulting in signal loss or attenuation.

Lowest Usable Frequency (LUF): The LUF is the lowest frequency that can be used for reliable communication between two specific locations via ionospheric reflection at a given time. It represents the lower limit of usable frequencies for a particular path under prevailing ionospheric conditions. Frequencies lower than the LUF may suffer from excessive absorption or attenuation in the ionosphere, making communication unreliable or impossible.

Optimum Working Frequency (OWF): The OWF is the frequency within the MUF and LUF range that provides the best signal strength and reliability for communication between two specific locations at a given time. It represents the frequency at which the radio waves experience the least absorption, reflection, or attenuation in the ionosphere, resulting in optimal transmission conditions. The OWF is often chosen to achieve the desired balance between range and reliability in HF communication.

The Frequency of Optimum Transmission (FOT) is typically 85 percent of the MUF. Generally, the FOT is lower at night and higher during the day

The ionosphere is a region of electrically charged particles or gases in the earth's atmosphere, extending from 50 to 600 km (approximately 30 to 375 miles) above the earth's surface.
There are layers of varying electron density in the ionosphere that absorb, pass, or reflect radio waves, depending on the density of the layer, the angle with which the radio waves strike it, and the frequency of the signal.
Ionization, caused by solar radiation, strips electrons from atoms, producing electrically charged particles.

SUMMARY

The density of ionospheric layers varies with the intensity of solar radiation, which changes according to time of day, season, and sunspot cycle.

The angle of radiation is determined by the wavelength of a signal and the type of antenna used.

Radio waves are absorbed as they pass through the ionosphere. The absorption rate increases as frequency decreases.

Communication is best at the frequency of optimum transmission (FOT), typically 85 percent of the maximum usable frequency (MUF).

Sunspots increase and decrease in 11-year cycles. Higher sunspot numbers increase ionization and therefore improve propagation. Lower sunspot numbers cause less ionization.

Solar flares cause sudden ionospheric disturbances (SIDs), which can disrupt HF communications for short periods of time.

Propagation prediction techniques, such as IONCAP and VOACAP, predict the MUF, LUF, and FOT for a given transmission path and time of day. Other methods include ionospheric sounding and Link Quality Analysis (LQA).

ELEMENTS of an HF Radio System

This section focuses on how they are generated

Basic primary components in a HF radio = transmitters, receivers, and antennas

Transmitters = exciters + power amplifiers

Exciter synthesizes a carrier wave that is modulated by a lower frequency signal derived from a source of information such as a microphone. Resulting signal is converted to the frequency to be transmitted.

Power amplifier boosts the output power of the signal to desired wattage for transmission before sending it through the RF cable to a transmitting antenna

Tunable Preselector/Bandpass filter removes noise, signals, and harmonics generated in the exciter or similar disruptions caused by the power amplifier

Receivers

Receivers include RF input filter/amplifier, frequency converters and immediate frequency amplifiers, a demodulator, and a local oscillator frequency synthesizer

RF Input Filter/Amplifier:

RF Input Filter: This is the first stage in the receiver chain. Its job is to filter out unwanted signals and noise from the desired radio frequency signal. This filtering is necessary to ensure that only the intended signal is processed by the receiver, reducing interference from other signals and improving the quality of the received signal.
Amplifier: After filtering, the signal is usually very weak and needs amplification. The amplifier increases the strength of the RF signal to a level suitable for processing by subsequent stages in the receiver.
Frequency Converters (Mixers) and Intermediate Frequency (IF) Amplifiers:

Frequency Converters (Mixers): This component mixes the incoming RF signal with a signal from a local oscillator to convert the RF signal to a lower, fixed intermediate frequency (IF). This process is known as heterodyning. The purpose of converting to an IF is to make the signal easier to filter and amplify in the later stages.

Intermediate Frequency (IF) Amplifiers: After conversion to IF, the signal is amplified further. IF amplification provides most of the gain in the receiver and improves the signal-to-noise ratio, which is crucial for clear signal reception.
Demodulator:

The demodulator is responsible for extracting the audio or data information from the carrier wave. This process is known as demodulation. The nature of the demodulator depends on the type of modulation used in transmitting the RF signal (e.g., AM, FM, QAM, etc.). The demodulated output is the intended audio or data signal.
Local Oscillator and Frequency Synthesizer:

Local Oscillator (LO): This component generates a stable frequency that mixes with the incoming RF signal in the mixer to produce the IF signal. The precision and stability of the local oscillator are crucial for accurate frequency conversion and, consequently, successful reception of the RF signal.
Frequency Synthesizer: This is often used to control the local oscillator frequency. A frequency synthesizer allows for precise and programmable frequency generation, enabling the receiver to tune to different frequencies and select different channels as desired.

In summary, the process involves initially filtering and amplifying the RF signal, converting it to an intermediate frequency for easier processing, demodulating it to extract the original audio or data content, and using a local oscillator and frequency synthesizer to accurately select and process the desired signal frequency.

Antennas

Some commonly used terms to describe antennas are impedance, VSWR, gain, radiation pattern, take-off angle, and polarization. Every antenna has an input impedance, which represents the load to be applied to the transmitter. This impedance depends upon many factors, such as antenna design, frequency of operation, and location of the antenna with respect to surrounding objects.

1. Impedance
Impedance in antennas refers to the resistance an antenna offers to the flow of electrical current and electromagnetic energy. It's measured in ohms (Ω). The input impedance of an antenna is crucial because it affects how efficiently the antenna can transfer radio frequency (RF) energy from the transmitter (or to the receiver). For maximum efficiency, the antenna's impedance should match the output impedance of the

transmitter (or the input impedance of the receiver). This matching minimizes reflections and maximizes power transfer.

2. VSWR (Voltage Standing Wave Ratio)
VSWR is a measure of how efficiently RF power is transmitted from the power source, through the transmission line, and into the load (in this case, the antenna). It's a ratio of the maximum to the minimum amplitudes of the standing wave pattern in the transmission line. A VSWR of 1:1 indicates perfect impedance matching, meaning all the power is transmitted to the antenna. Higher ratios indicate more significant mismatches, leading to more power being reflected back towards the transmitter, which is inefficient and can even damage the transmitter.

3. Gain
Gain of an antenna is a measure of its ability to direct or concentrate RF energy in a particular direction compared to a standard reference antenna (usually an isotropic radiator, which radiates equally in all directions). Gain is typically measured in decibels (dB). A higher gain indicates a more focused beam of RF energy, which can increase the effective range of the antenna. However, it's important to note that antennas don't create energy; they just redistribute it. So, increasing gain in one direction comes at the expense of reduced radiation in other directions.

Gain can be expressed in dBi; the higher this number, the greater the directivity of the antenna. Transmitting antenna gain directly affects transmitter power requirements. If, for example, an omnidirectional antenna were replaced by a directional antenna with a gain of 10 dBi, a 100 Watt transmitter would produce the same effective radiated power (ERP) as a 1 kW transmitter and omnidirectional antenna.

4. Radiation Pattern
The radiation pattern of an antenna describes how it distributes energy into space. It's usually represented graphically, showing the variation of power radiated or received by the antenna at different angles around it. This pattern helps in understanding the antenna's directional characteristics – where it radiates or receives energy most effectively and where it has nulls or minimal radiation.

5. Take-off Angle
The take-off angle is particularly relevant for high-frequency (HF) antennas used in long-distance communications. It refers to the angle between the RF wave leaving the antenna and the Earth's surface. A lower take-off angle is generally preferred for long-distance communications because it allows the signal to reach the ionosphere at a sharper angle, facilitating better long-range skip propagation.

6. Polarization
Polarization refers to the orientation of the electric field of the electromagnetic wave radiated by the antenna. Common types of polarization include vertical, horizontal, circular, and elliptical. The receiving antenna must have the same polarization as the transmitting antenna for optimal reception. Mismatched polarizations can lead to signal loss.

Basic challenge in radio communications = finding ways to get the most power, where and when you need it, to generate and transmit signals

Most transmitters are designed to provide maximum output power and efficiency into a 50 Ohm load.

Ohms measure resistance

Log periodic/broadband antennas: can provide a 50 Ohm load to the transmitter over a wide range of frequencies. These antennas can generally be connected directly to the transmitter.

1. Dipole Antennas
Dipole Antennas are among the simplest and most widely used types of antennas. A basic dipole consists of two conductive elements (usually metal wires or rods) arranged end-to-end with a small gap in the middle where it is connected to the transmission line. Dipoles are often described as "half-wave" dipoles because their total length is typically half the wavelength of the frequency they are designed to operate at. The dipole antenna radiates RF energy efficiently and has a radiation pattern resembling a figure-eight, with the maximum radiation perpendicular to the antenna.
2. Whip Antennas
Whip Antennas are a type of monopole antenna and can be considered a half-dipole. They consist of a single conductive rod, often flexible, mounted perpendicularly over a ground plane (which can be the earth, a vehicle roof, or a metal surface). Whip antennas are simple, rugged, and easy to use, making them popular for portable and vehicle-mounted applications (like car radios and handheld transceivers). They generally have an omnidirectional radiation pattern, radiating RF energy equally in all directions in one plane.
3. Long-Wire Antennas
Long-Wire Antennas are a simple and effective design used for long-distance communication, especially in HF bands. As the name suggests, these antennas consist of a single, long wire, often multiple wavelengths long, making them "long" relative to the wavelength of the frequency they are intended to use. The radiation pattern of a

long-wire antenna becomes more directional and complex as the length increases, often favoring certain directions over others, which can be advantageous for long-distance communications.

Why a Coupler/Tuner is Needed
Impedance Matching: One of the primary reasons for using a coupler or tuner with these antennas is to match their impedance with that of the transmitter or receiver. Mismatches in impedance can lead to reflections of the RF energy back toward the transmitter, reducing efficiency and possibly damaging the transmitter's output stage.

Bandwidth Enhancement: Many antennas, especially dipoles and long-wires, have a relatively narrow bandwidth where they perform optimally. A tuner can help adjust the effective electrical length of the antenna, making it perform better outside its natural resonant frequency band.

Versatility: With a tuner, an antenna can be used across a broader range of frequencies than would otherwise be possible. This is particularly useful for long-wire antennas, which might not be resonant at all desired frequencies of operation.

Convenience: In practical applications, it's not always possible to have a perfectly sized antenna for every frequency of interest, especially in portable or mobile setups. A tuner allows for the flexible use of a single antenna across various frequencies, which is especially valuable in situations where changing antennas is impractical.

Radiation pattern is determined by an antenna's design and, in the HF band, the pattern is strongly influenced by its location with respect to the ground. It may also be affected by its proximity to nearby objects such as buildings and trees. In most antennas, the pattern is not uniform, but is characterized by lobes (areas of strong radiation) and nulls (areas of weak radiation).

take-off angle is the angle between the main lobe of an antenna pattern and the horizontal plane of the earth's surface.

Low take-off angles are generally used for long-haul communications; high take-off angles are used for shorter-range communications.

One of the most versatile types of HF antenna is the half-wave dipole, which is basically a length of wire equal to one-half the transmitting wavelength and excited in the center. The dipole can be oriented to provide either horizontal or vertical polarization

Sky wave communications between relatively closely spaced stations may require antennas specially designed for this purpose. These near vertical incidence sky wave (NVIS) antennas have a very high take-off angle, radiating RF energy nearly straight up. The radio waves refract downward to the earth in a circular pattern. NVIS antennas provide omnidirectional coverage out to about 400 km.

SUMMARY

A radio system consists of a transmitter, receiver, and antenna group.

The transmitter group consists of an exciter and power amplifier. The exciter includes a modulator, carrier generator, and frequency transla- tor.

The receiver group consists of an RF input filter/amplifier, frequency converters/IF amplifiers, demodulator, and local oscillator.

Antenna selection is critical to successful HF communications. Antenna types include vertical whip, dipole, and directional.

An antenna coupler matches the impedance of the antenna to that of the transmitter, transferring maximum power to the antenna.

The gain of an antenna is a measure of its directivity — its ability to focus the energy it radiates in a particular direction.

Antenna radiation patterns are characterized by nulls (areas of weak radiation) and lobes (areas of strong radiation).

Low antenna take-off angles are generally used for ground-wave and long-haul communications; high take-off angles are used for shorter-range

External noise levels greatly exceed internal receiver noise over most of the HF band. Signal quality is indicated by signal-to-noise ratio (SNR), measured in decibels (dB). The higher the SNR, the better the signal quality. Interference may be inadvertent, as in the case of the pilot's call to the tower. Or, it may be a deliberate attempt on the part of an adversary to disrupt an operator's ability to communicate

To combat noise and interference, techniques involve....
(1) boosting the effective radiated power, (2) providing a means for optimizing operating frequency, (3) choosing a suitable modulation scheme, (4) selecting the appropriate antenna system, (5) designing receivers that reject interfering signals, and (6) designing data waveforms that function in very poor noise and interference envi- ronments

Types of Interference

Lightning is the main atmospheric (natural) source of noise. Atmospheric noise is highest during the summer and greatest at night, especially in the 1.6 to 5 MHz range. Another natural noise source is galactic or cosmic noise, generated in outer space. It is uniformly distributed over the HF spectrum, and dominates the noise floor above 20 MHz.

Power lines, computer equipment, and industrial and office machinery produce man-made noise, which can reach a receiver through radiation or by conduction through power cables. This type of man-made noise is called electromagnetic interference (EMI) and it is highest in urban areas. Grounding and shielding of the radio equipment and filtering of AC power input lines are techniques used by to suppress EMI.

HF transmitters compete for space on the radio spectrum in a relatively narrow range of frequencies, causing interference with one another. Interference is most severe at night in the lower bands at frequencies close to the MUF. The HF radio spec- trum is especially congested in Europe due to the density of the population.

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Fading is a phenomena caused by the reality of multiple paths existing to reach a receiver. Fading is a variation in average signal level because signals may add or subtract from each other in a random way. Engineers avoid the effects of multipath and fading by designing waveforms resistant to this phenomena

Deliberate interference, or jamming, results from transmitting on oper- ating frequencies with the intent to disrupt communications. Jamming can be directed at a single channel, multiple channels, or wider band- widths. It may be continuous (constant transmitting) or look-through (transmitting only when the signal to be jammed is present).

Three techniques are employed to defeat this threat: spread spectrum, tone excision, and frequency hopping. Spread-spectrum techniques cause the modulated information to be transmitted in a bandwidth consider- ably greater than the frequency content of the original information. Software-based tone excision techniques are applied with some modems to eliminate interfering tones, some as high as 40 dB above the signal. Frequency hopping is a special mode of a radio in which all radios in the net change frequencies quickly and synchronously to avoid intentional jamming.

Modems (from MOdulator/DEModulator)

Modems, which stand for "modulator-demodulator," are devices used to transmit digital data over a communication line that typically carries analog signals, such as a telephone line. They work by modulating digital signals into analog for transmission and demodulating received analog signals back into digital form.

The application of error-correcting codes and automatic repeat request (ARQ) techniques offering error-free data transfer permits the use of HF radio in computer-to-computer communications systems

Communication involves the transfer of information from a transmitter to a receiver over a suitable channel

Book analogy to understand data transmission over the air

It uses symbols (the alphabet) to encode information into a set of code groups (words) for transmission over a channel (the printed page) to a receiver (the reader). Applying this principle to data transmission, we begin by using a kind of shorthand to transform the data into code words (binary digits, or bits) for transmission over a channel, (HF radio) to a receiver, (the operator).

Data transmission speed is commonly measured in bits per second (bps).
The maximum bit rate that can be supported by a radio channel depends on its bandwidth — the greater the bandwidth, the greater the bit rate.

Synchronous and asynchronous data transmissions are two primary methods used to transmit data. Most high speed HF modems are used in synchronous data mode.

Synchronous Data Transmission

- In synchronous transmission, data is sent in a continuous stream, usually accompanied by timing signals. These timing signals are used to synchronize the transmitter and receiver's clocks to ensure that the receiver can correctly interpret the incoming bits.
- Data is organized into frames or blocks of a certain size, and these frames are sent continuously at a regular interval defined by the clock signal.
- It often involves a dedicated communication channel between the sender and receiver.

- Efficient for transmitting large volumes of data because the overhead of start and stop bits (as in asynchronous transmission) is not required for each byte or character.
- The constant stream of data and the presence of clock signals allow for faster data rates compared to asynchronous transmission.

- Requires more complex hardware and setup to maintain the clock signal and synchronization between the sender and receiver, which can increase the cost.
- Less flexible in accommodating devices with varying data rate requirements or devices that do not constantly send or receive data.

Asynchronous Data Transmission

In asynchronous transmission, data is sent one byte or character at a time, and each byte is framed with start and stop bits. The start bit indicates the beginning of a byte, and the stop bit(s) indicate the end.

There is no need for the sender and receiver to have synchronized clocks because the timing of each byte is independent, and the presence of start and stop bits allows the receiver to identify each byte.

Often used for communication over short distances or where data is sent sporadically.

Simpler and cheaper to implement than synchronous transmission because it does not require clock synchronization between the sender and receiver.

More flexible in accommodating devices with different data rate requirements or devices that send data intermittently.

Less efficient for transmitting large volumes of data due to the overhead of start and stop bits for each byte, which reduces the effective data rate.

Generally slower than synchronous transmission because of the additional bits required for each byte and the potential gaps between characters.

Synchronous transmission is more efficient for large data volumes due to continuous streaming and the lack of start/stop bits for each byte. Asynchronous transmission, while less efficient for large data volumes, is more suited for sporadic or low-volume communication.

Synchronous systems are more complex and costly due to the need for clock synchronization. Asynchronous systems are simpler and cheaper, making them more accessible for simple or small-scale applications.

Asynchronous transmission is more flexible in handling data from devices with different speeds or intermittent data flows. Synchronous transmission, while less flexible, is ideal for consistent, high-speed data flows.

Synchronous transmission is often used in high-speed, large-volume data transfer scenarios such as between computer components or in high-speed network connections.

Asynchronous transmission is common in *serial communication, such as USB, keyboard, and mouse connections, where data volume is low and sporadic.*

Most older voice radios cannot transmit data directly. Data digital voltage levels must be converted to audio, using a device called a modula- tor, which applies the audio to the transmitter. Conversely, at the receiver, a demodulator converts audio back to digital voltage levels. Newer radios are equipped with built-in high-speed modems which permit the radios to operate with either voice or data inputs.

HF modems fall into three basic categories: (1) modems with slow- speed frequency shift keying (FSK); (2) mid-rate 2400 bps modems; and (3) high-rate 9600 bps modems.

The comparison among slow-speed modems with Frequency Shift Keying (FSK), mid-rate 2400 bps modems, and high-rate 9600 bps modems can be understood in terms of their modulation techniques, data rates, and typical applications.

1) Modems with Slow-Speed Frequency Shift Keying (FSK)

Modulation Technique: FSK is a form of frequency modulation where binary data is represented by two different frequencies; one frequency represents a binary 1, and the other represents a binary 0. Slow-speed modems using FSK are relatively simple in design.

Data Rate: These modems typically operate at very low data rates, usually up to 300 bps (bits per second).

Applications: Given their low data rate, they were commonly used in early telecommunications applications, such as teletypewriter (TTY) systems and the earliest computer modems for dial-up Internet access.

2) Mid-Rate 2400 bps Modems

Modulation Technique: Mid-rate modems often use more complex modulation techniques than FSK, such as Phase Shift Keying (PSK) or Quadrature Amplitude Modulation (QAM). These techniques are more efficient in how they use the available bandwidth, allowing for higher data rates.

Data Rate: As indicated, these modems can transmit data at rates up to 2400 bps. This was considered a significant improvement over the earlier, slower modems and was suitable for early internet access and fax transmission.

Applications: Mid-rate modems were popular in the late 1980s and early 1990s for personal and business use, enabling more practical internet access, file transfers, and facsimile (fax) transmissions over standard telephone lines.

3) High-Rate 9600 bps Modems

Modulation Technique: High-rate modems like the 9600 bps models use even more sophisticated modulation techniques, often incorporating multiple bits per symbol to increase the data rate significantly. Techniques such as higher-order QAM are common, which combine both amplitude and phase modulation to represent multiple bits with each symbol.

Data Rate: These modems are capable of transmitting data at rates up to 9600 bps, which was quite fast for dial-up internet standards and significantly improved the user experience by reducing download times and improving web browsing speeds.

Applications: High-rate modems were widely used in the 1990s for internet access, particularly in business and academic settings where faster data transmission was necessary. They facilitated more advanced online applications, including the early stages of multimedia content delivery.

Compare and Contrast

- Modulation Techniques: The primary difference lies in the modulation techniques used, with FSK being the simplest and suitable for low data rates, while PSK, QAM, and higher-order variations thereof allow for higher data rates by more efficiently using the available bandwidth.
- Data Rates: Data rates increase from FSK-based slow-speed modems (up to 300 bps) to mid-rate (2400 bps) and high-rate modems (9600 bps), reflecting advancements in modulation technology and the increasing demand for faster data transmission.
- Applications: Slow-speed modems were suitable for basic text transmission and early computer communications. In contrast, mid-rate modems expanded the possibilities to include internet access and fax, and high-rate modems further enhanced internet experiences and supported more data-intensive applications.

Serial tone modems provide vastly improved data communications on HF channels, including a higher data rate with powerful forward error correction (FEC), greater robustness, and reduced sensitivity to interfer- ence.

FEC systems provide error correction without the need for a return link.

Interleaving is a technique that randomizes error bursts, allowing FEC systems to work more effectively.

Soft-decision decoding further reduces bit error rates by comparing a group of symbols that retain their analog character against the set of possible transmitted code words.

Automatic channel equalization and adaptive excision filtering are sig- nal processing techniques that improve data communications performance

ALE Technology

ALE is a technique that permits HF radio stations to call and link on the best HF channel automatically. Typically, ALE systems make use of recently measured radio channel characteristics (LQA data) stored in a memory "matrix." The system works much like a telephone in that each radio in a network is assigned an address. When not in use, each radio receiver constantly scans through its assigned frequencies, listen- ing for calls addressed to it.

To reach a specific station, the caller simply selects an ALE address, just like dialing a phone number. The radio consults its LQA matrix and selects the best available assigned frequency. It then sends out a brief message containing the destination address. When the receiving sta- tion "hears" its address, it stops scanning and stays on that frequency. The two stations automatically conduct a "handshake" to confirm that a link is established and they are ready to communicate

The receiving station, which was completely silent while scanning, will typically emit a ringing signal to alert the receiving operator of an incoming call. At the conclusion of the call, one of the stations "hangs up" a disconnect signal is sent to the other station, and they each return to the scanning mode. If the units don't link on the first select- ed channel, the system selects the next best channel from the LQA matrix and calls. This process is repeated until the systems link, or the channel list is exhausted.

More on LQA

At programmed intervals, a station in a network will attempt to link with another station on each of its assigned frequencies and measure the signal quality. These quality scores are stored in a matrix. When a call is initiated, the radio checks its "LQA matrix" to determine the best frequency to call the desired station. It then attempts to link on that frequency. If the link cannot be established, it will try again on the next best frequency, and so on, until a link is established.

When making multi-station calls, the radio selects the channel with the best average score.

The difference between an Exchange and a Sound is that an Exchange is a bi-directional test, whereas a Sound is a one-way broadcast. Exchanges are typically used when there is regular traffic between two specific points. Sounds are often used where there is a central site that all “outstations” call into or in larger networks.

Link Protection

Link protection (LP) is the scrambling of ALE signaling. In strict terms, it is not a COMSEC or TRANSEC function and it often called LINKSEC. Link Protection has two purposes: anti-spoofing and address protec- tion. An ALE system is vulnerable to spoofing from an automated sys- tem or simply from a tape recorder. For example, an adversary could simply record a link termination transmission to your system off-the-air and play it back over-the-air when your system is linked. Your system would properly interpret the transmission as a “return to scan” mes- sage and do just that. Secondly, if address information is sent in the clear, an adversary could know both your address and the address of the station you are talking to.

Link Protection addresses both these issues by scrambling the on-air ALE bits. All units in a net have the LP key and can decode the incom- ing signals. The scrambling changes periodically (every “Protection Interval”) to defeat the tape recorder attack.

Digital Voice

The first digital voice systems were designed to provide secure voice.

LPC-10e based vocoders were standardized in the 80s and 90s and were fielded in large numbers.

The higher data rate of vocoders have better voice quality than lower rate vocoders of their same type, and have lower end-to-end delays.

600 bps vocoders provide significantly better reliability than 2400 bps.

MELP vocoders have been fielded recently and provide enhanced voice quality and intelligibility.

600 bps MELP Vocoders provide voice quality superior to that of 2400 bps LPC-10e vocoders, with greatly improved reliability due to the lower data rate

Internet Protocol (IP): It is relatively simple to send IP-based traffic over HF. The difficulty lies in getting IP systems to work well over HF radios. There are several obstacles to overcome because IP applications are generally:

+ Designed for reliable links.
+ Designed for wired data rates.
+ Expecting low latency.
+ Chatty (including TCP retries).

These problems are overcome by a combination of 3G and radio soft-
ware features. Networked military software applications use a combi- nation of reliable (TCP-based) and unreliable (UDP-based) protocols. In either case, the applications need nearly error-free data to arrive at the destination to work effectively. A 3G IP radio will utilize the xDL proto- col suite to insure that the data arrives error-free. This is transparent to the application protocol stack and is much more efficient than relying on TCP for error correction. The HF radio will use a combination of techniques to improve the data rate including utilization of the fastest of the data modes (e.g. HDL+), compression of each transmission, and elimination of all redundant retry packets sent by TCP

STANAG 5066 and 4538 are still viable and continually updated with new “editions”

Securing HF Communications

There are four things an enemy would like to do with any transmitted information:
1. The enemy wants to know when information is being transmitted.
(TRANSEC Concern)
2. The enemy wants to intercept the information. (TRANSEC Concern) 3. The enemy wants to be able to understand and use the information.
(COMSEC Concern)
4. If the enemy cannot understand the information, then they would
want to stop the transmission of information. (TRANSEC Concern)
There are two key methods to prevent the enemy from succeeding: COMSEC and TRANSEC.

COMSEC
COMSEC uses scrambling or cryptographic techniques in order to make information unintelligible to people who do not have a need to know or who should not know. We'll differentiate here between crypto- graphic or ciphering techniques applied to digital signals and scram- bling techniques applied to analog signals.

Cryptography is the process of encrypting (translating) information into an apparently random message at the transmitter and then deciphering the random message by decryption at the receiver

In digital encryption the data, which may be digitized voice, is reduced to a binary data stream. The cryptographic engine creates an extremely long, non-repeating binary num- ber stream based on a traffic encryption key (TEK). The data stream is added to the cryptographic stream, creating the encrypted data, or cipher text. A binary stream created in this fashion is inherently unpre- dictable; it also provides a very secure method of protecting informa- tion. On the other hand, all analog signals are more predictable and

In voice communications systems that do not require extremely high security, you can protect against casual eavesdropping by scrambling. Scrambling, as an analog COMSEC technique, involves separating the voice signal into a number of audio sub-bands, shifting each sub-band to a different audio frequency range, and combining the resulting sub- bands into a composite audio output that modulates the transmitter. A random pattern controls the frequency shifting. The technique of scrambling the pattern is similar to sending a message with a decoder ring, like the ones sometimes found in children's cereal boxes.

You can, for example, designate that the letter c be ciphered as g, a as n, and t as w, so that when you receive the message gnw, you decode it as cat.

Descrambling occurs at the receiver by reversing the process. Harris' Analog Voice Security (AVS) allows for easy entry into the communica- tions net because it doesn't require synchronization with other stations.

Symmetric and asymmetric encryption are two fundamental methods used to encrypt data, each with its unique characteristics and use cases.

Symmetric Encryption:
- Utilizes a single key for both encryption and decryption of data.
- Known for its speed and efficiency due to less complex algorithms and shorter key lengths compared to asymmetric encryption.
- Common symmetric encryption algorithms include AES (Advanced Encryption Standard), DES (Data Encryption Standard), and 3DES (Triple DES).
- Ideal for applications where large volumes of data need to be encrypted, such as file storage or database encryption.
- The major challenge with symmetric encryption is the secure exchange of the encryption key between parties, as anyone with access to the key can decrypt the data

Asymmetric Encryption:
- Employs a pair of keys for encryption and decryption: a public key, which is shared openly for encrypting data, and a private key, which is kept secret and used for decryption.
- Asymmetric encryption provides enhanced security features, such as digital signatures, which authenticate the identity of the sender and ensure the integrity of the message.
- Well-known asymmetric encryption algorithms include RSA, Diffie-Hellman key exchange, and ECC (Elliptic Curve Cryptography).
- This form of encryption is particularly suitable for secure communications over untrusted networks, like the internet, and for establishing secure connections where identity verification is crucial.
- The downsides include slower performance due to longer key lengths and more complex algorithms, and the potential for security breaches if the private key is lost or compromised

In practice, both encryption types are often used together in various security protocols to leverage their respective strengths. For instance, asymmetric encryption is used to exchange symmetric encryption keys securely over an untrusted network, combining the efficiency of symmetric encryption with the secure key exchange mechanism of

asymmetric encryption. This hybrid approach is commonly seen in SSL/TLS protocols for secure web browsing.

Transec

The most commonly used TRANSEC technique is frequency hopping. In this system, the transmitter frequency changes so rapidly that it is difficult for anyone not authorized to listen in or to jam the signal. The receiver is synchronized so that it hops from frequency to frequency in a predetermined pattern in unison with the transmitter. Frequency hop- ping scatters the intelligence over several hundred discrete frequencies. A radio operator listening to one of these frequencies may hear a short "pop" of static. A broadband receiver could perhaps capture all of these little bursts; however, the task of picking these bursts out of the other natural and man-made bits of noise would be daunting, requir- ing a team of experts several hours just to reassemble a short conver- sation. Jamming one channel would have minimal impact on the hop- ping communicator. To effectively jam a frequency-hopping radio, most or all of the frequencies that the hopping communicator uses would have to be jammed, thus preventing the use of those frequencies by the adversary as well.

SUMMARY

COMSEC uses cryptography or scrambling to make information unintelligible to people who do not have a need to know.

+ The security level of a COMSEC system depends on the mathematical soundness of the algorithms and the number of variables in the key.

+ Protecting the key is vital to securing the transmitted information. + Public key cryptography is widely used in the commercial sector.

Over-The-Air-Rekeying (OTAR) eliminates the need for manual loading of keys and provides a more secure method of key management.

TRANSEC protects the transmitted signal itself, to prevent signal detec- tion or jamming of the transmission path.

+ Low Probability of Detection (LPD) systems use spread-spectrum and other techniques to "hide" the signal beneath the natural noise level.

+ Low Probability of Interception (LPI) radios transmit compressed digital data in short bursts or over a wide bandwidth.

+ Frequency-hopping radio systems jump rapidly in unison, from one frequency to another in apparently random patterns, using a

VHF/UHF Communications

VHF: Man to Man The VHF band was an early choice for manpack radios used by ground troops to communicate within a local (five-mile or so) area. Antennas and selective-tuning components of VHF radios are much smaller than their HF counterparts. Advances in the semiconductor industry have also increased the efficiency of VHF radios because batteries are smaller, lighter, and longer-lived than those required in the past. Unlike HF, VHF transmissions lack the ability for ionospheric bounce and are limited to line-of-sight (LOS) communication. This reduces radio emission clutter throughout an extended battlefield and limits the vulnerability to unfriendly interception. The wider channel bandwidth capabilities of VHF radios increase the efficiency of coding and encryption schemes and allow greater data throughput than that of HF radios. Wider bandwidth and limited range make these radios ideal for squad-to-squad communications.

UHF: Ground to Air for Close Support
UHF tuning elements and antennas are even smaller than those of VHF and are much easier to mount on supersonic fighter aircraft, making UHF an ideal choice for ground-to-air communications. Like VHF radios, UHF radios share the advantages of being line of sight and having wide bandwidth. Modern military forces now prefer the UHF spectrum for ground-to-air communications.

Comparison of UHF and VHF…
VHF (Very High Frequency)
Frequency Range: VHF covers from 30 MHz to 300 MHz.
Wavelength: Wavelengths in the VHF band range from about 1 meter to 10 meters.
Propagation: VHF signals primarily travel by line-of-sight, meaning they do not significantly bend over the horizon or reflect off the ionosphere (except under certain atmospheric conditions, like temperature inversions). However, their longer wavelengths can diffract over and around obstacles to some extent, providing slightly better coverage in hilly or forested terrain compared to UHF.
Applications: Due to its relatively longer wavelength, VHF is well-suited for outdoor communications over moderate distances, such as marine communications, two-way land mobile radios (e.g., services used by police, fire, and ambulance), and FM broadcasting.
Antenna Size: The antennas for VHF are typically larger than those for UHF due to the longer wavelength. This can be a consideration for portable devices.

UHF (Ultra High Frequency)
Frequency Range: UHF covers from 300 MHz to 3 GHz.
Wavelength: Wavelengths in the UHF band range from approximately 1 meter to 10 centimeters.
Propagation: UHF signals are also primarily line-of-sight but have a shorter wavelength, which allows them to penetrate through urban environments more effectively. However, they are more attenuated by obstacles and tend not to diffract around objects as well as VHF, limiting their range in areas with many obstructions.
Applications: UHF is extensively used for TV broadcasting, cellular networks, GPS, Bluetooth, Wi-Fi, and walkie-talkies, especially in urban and densely populated areas. Its shorter wavelengths allow for smaller antennas, making it ideal for handheld and vehicle-mounted devices.
Antenna Size: UHF antennas are smaller and more compact, which is advantageous for portable and mobile applications.

Key Differences and Considerations
Both VHF and UHF have line-of-sight propagation, but their interaction with various environments differs due to their wavelengths. VHF can provide slightly better range in open or slightly obstructed environments, while UHF is better suited for densely built-up areas.

UHF tends to have more available spectrum and is heavily used for many modern digital communication systems, which can lead to congestion in certain bands.

LOS is typically 5 miles or less along the ground between manpack radios

Vertical LOS range of UHF is tens of thousands of miles, enabling it to reach satellite and allow for satellite communication (SATCOM). This is thanks to the use of highly directional antennas pointed up at the sky (vertically polarized).

TACSAT = Tactical Satellite

The Special Forces combat teams are very different. Although they need to have access to all of the available military communication channels, they can't afford the luxury (or the weight) of carrying the required multitude of radios. Hence, the multiband radio.

Multiband systems can operate across multiple frequency bands, offering the capability to access different ranges of frequencies as needed. This doesn't necessarily mean the device can use multiple bands simultaneously; rather, it can switch between them according to user requirements or predefined settings. Multiband capabilities are particularly valuable in devices that need to operate in diverse environments or for different communication standards. For example, a multiband radio could access VHF frequencies for marine communications and UHF frequencies for urban or indoor environments. The primary advantage of multiband systems is their versatility. They can adapt to different communication needs, regulatory environments, and operational conditions by switching between frequency bands. This makes them ideal for users who need to communicate over different ranges or in varied scenarios, such as emergency services, military operations, and international travelers.

NOTE:
A band is a specific range of frequencies within the radio spectrum, defined by its lower and upper frequency limits. Bands are broad segments of the spectrum allocated for various uses, such as broadcasting, mobile communications, Wi-Fi, and satellite communications. For example, the VHF (Very High Frequency) band spans from 30 to 300 MHz, and within this band, there are specific allocations for different types of services like FM radio, television, and marine communications.

A channel, on the other hand, is a specific part of a band that is designated for a particular use or service. Channels are essentially slices of a band, each with a defined width (bandwidth) and center frequency. For instance, in the FM radio band, each station is assigned a specific channel within the broader FM band (typically 88 to 108 MHz in many regions). These channels are usually spaced apart to prevent interference, allowing multiple stations to operate within the same band without overlapping.

A band covers a wider range of frequencies and encompasses many potential channels, while a channel refers to a narrower frequency range within a band. Bands are generally defined to organize the spectrum into regions for particular types of communication or technology. Channels are designated within these bands to facilitate specific services or transmissions. Within a given band, there can be flexibility in how channels are defined and used, depending on regulatory decisions and the specific requirements of the services operating within that band.

Reiterating the HF, VHF, UHF Bands

The High Frequency (HF) Band
The HF band is defined as the frequency range of 3 to 30 MHz. In practice, most HF radios use the spectrum from 1.6 to 30 MHz. Most long-haul communications in this band take place between 4 and 18 MHz. Higher frequencies (18 to 30 MHz) may also be available from time to time, depending on ionospheric conditions and the time of day (see Volume One, HF Technology).

Very High Frequency (VHF) Band
The VHF frequency band is defined as the frequency range from 30 to 300 MHz. From the previous discussion about the relationship between frequency and wavelength, it should be noted that VHF wavelengths vary from 10-meters at the low end to one meter at the high end. This means that the size of antennas and tuning components used in VHF radio are much smaller and lighter than those of HF radios. This is a big advantage for manpack radios. We will also see in later chapters that the higher frequency and shorter wavelengths of VHF radios have a profound effect on radio range.

Ultra High Frequency (UHF) Band
The UHF band goes from 300 MHz to 2450 MHz, although TACSAT manpack UHF radios do not utilize frequencies above 512 MHz. The wavelengths associated with 300 to 512 MHz range from one meter to 0.58 meters (58 centimeters). The very small antennas required for these wavelengths make them ideal for use on high-speed aircraft.

Frequency Allocation → Type of Service + Type of Transmission

Carrier Waves → Radio waves modulated to carry information

Phase → measurement of a complete wave cycle

Simplest way of modulating a carrier → On-off keying using morse code

Amplitude Modulation → creates a carrier and pair of duplicate side bands

Single Side Band → Carrier Wave and one of the side bands are suppressed and a singular side band is transmitted

Amplitude modulation equivalent (AME) → a

carrier at a reduced level is transmitted with the sideband. AME lets one use a relatively simple receiver to detect the signal

Independent sideband (ISB) → both an upper and lower sideband, each carrying different information, is transmitted. This way one sideband can carry a data signal and the other can carry a voice signal.

Frequency Modulation → the carrier's frequency varies in response to changes in the modulating signal and uses substantially more bandwidth than amplitude modulation

Bandwidth = Upper Frequency Limit - Lower Frequency Limit

HF Bandwidth 28.4 MHz

VHF Bandwidth 270 MHz

Long-range radio communications beyond line of sight (BLOS) take place in the high-frequency (HF) range of 1.6 to 30 MHz. Different portions of this band are allocated to specific radio services under international agreement.

Short-range radio communications (LOS) communications can take place at all radio frequencies, but that task is most often given to the VHF and UHF radio bands.

Sky wave propagation is generally not available in the VHF and UHF frequency bands.

Modulation is the process whereby the phase, amplitude, or frequency of a carrier signal is modified to convey information

Visible Horizon (at five feet above flat surface → less than 2.7 miles

This is the max LOS radio range from manpack radio to another

Height of transmitting and receiving antennas is important to determine distance of effective data transmission

For most systems, when doubling the distance, the radiated signal is divided by four! Therefore, transmit power is often the limiting range factor. It is common to see 500-watt and 1-kW HF transmitters in vehicular or shipboard HF applications, and 10-kW or greater for HF fixed station broadcast sites.

Multipath propagation is a phenomenon in radio communications where signal waves travel from the transmitter to the receiver via multiple paths. This occurs due to the reflection, diffraction, and scattering of the radio waves by various obstacles such as buildings, terrain features, and atmospheric layers. Multipath propagation is common in urban environments, indoor settings, and hilly or mountainous areas.

Effects of Multipathing

Signal Fading: Multipath can lead to constructive and destructive interference as the multiple signal paths combine at the receiver. Destructive interference, where signal waves cancel each other out, can cause signal fading, reducing the signal's strength and quality.

Multipath Distortion: When different paths cause parts of the signal to arrive at slightly different times, it can lead to multipath distortion. This is particularly problematic for digital signals, where timing is crucial, as it can cause intersymbol interference (ISI) where one symbol interferes with subsequent symbols, making it difficult for the receiver to correctly interpret the signal.

Phase Shifts: Multipath can also introduce phase shifts in the signal, further complicating its reception and decoding.

VHF/UHF Transceivers

The simplest transceiver must generate a modulated signal to the antenna and to receive a signal from an antenna, demodulate it, and feed the information to a headset, computer, or some other human or machine interface. A multiband transceiver must perform these functions for each of its frequency bands.

A radio system consists of a transceiver and an antenna group.
The transceiver provides both transmitting and receiving functions.
The transmit function consists of modulation, carrier generation frequency translation, and power amplification.
The receive function consists of RF signal filtering, amplification, frequency down conversion, and demodulation.

Antenna selection is critical to successful VHF, UHF, and TACSAT communications.
Antenna types include vertical whips, center-fed dipoles, biconical antennas, directional log periodic arrays, and umbrella TACSAT antennas.
An antenna coupler matches the impedance of the antenna to that of the transmitter, transferring maximum power to the antenna.
The gain of an antenna is a measure of its directivity — its ability to focus the energy it radiates in a particular direction.
Antenna radiation patterns are characterized by nulls (areas of weak radiation) and lobes (areas of strong radiation).

Interference
Natural (atmospheric) and man-made sources cause noise and interference. Power lines, computer terminals, and industrial machinery are prominent causes of man-made noise. Congestion of radio transmitters competing for limited radio spectrum causes interference.
Collocated transmitters interfere with nearby receivers.
Jamming, or deliberate interference, results from transmitting on operating frequencies with the intent to disrupt communications.
Multipath interference can be considered another form of noise.
Proper antenna selection and advanced modulation techniques can reduce the effects of noise and interference.

Baud → a unit of signaling speed and is a measure of symbols per second that are being sent. A symbol may represent more than one bit.

Data communications techniques are also used for encrypting voice calls by a device called a vocoder (short for voice coder- decoder). The vocoder converts sound into a data stream for transmission over an HF radio channel. A vocoder at the receiving end reconstructs the data into telephone-quality sound.

The transmission of data requires the use of modems to convert digital data RF signal form when transmitting, and convert the RF signal back to digital form when receiving.
Radio modems are classified as slow-speed FSK, high-speed parallel tone, or high-speed serial tone.
Serial tone modems provide vastly improved data communications, including a higher data rate with powerful forward error correction (FEC), greater robustness, and reduced sensitivity to interference.
FEC systems provide error correction without the need for a return link.

Interleaving is a technique; mostly used for HF channels that randomizes error bursts, allowing FEC systems to work more effectively.
Soft-decision decoding further reduces bit error rates by comparing a group of symbols that retain their analog character against the set of possible transmitted code words.
A vocoder converts voice signals into digital data for coded transmission over HF channels.
Automatic channel equalization and adaptive excision filtering are signal processing techniques that improve data communications performance.
M-ary Phase Shift Keying is a method of increasing the data rate of radio transmissions. "M" refers to the number of phases used in the modulation scheme.
Trellis Coded Modulation (TCM) is a coding technique that provides maximum data rate capability to PSK data streams by improving the noise margin.

UHF SATCOM

A Communications Satellite is a Radio Repeater!!!!

SATCOM transceivers achieve coverage by using radio repeaters placed in satellites.

The advantage that antenna height makes in extending LOS distance is extensive. A satellite is the ultimate high antenna tower. UHF is an excellent candidate frequency to contact a satellite because it can penetrate the atmosphere and ionosphere with little attenuation.

Directional antennas are used as SATCOM antennas, meaning they are constructed with a reflector similar to those used in a flashlight to focus the beam. By focusing the beam of a directional antenna, you can boost the effective radiated power by four times or more.

DAMA (Demand Assigned Multiple Access) is a communication technique used in satellite, radio, and other wireless communication systems. It is designed to efficiently allocate channel capacity among multiple users based on demand, rather than assigning fixed channels to users regardless of their current need for the channel. This dynamic allocation of bandwidth and channel resources makes DAMA an effective method for managing communication in networks where the number of users can vary significantly, and the usage pattern is unpredictable.

How DAMA Works

Request for Channel: When a user needs to transmit data, their terminal sends a request to the network controller (which can be a satellite, a base station, or another type of central controller) for a communication channel.

Channel Assignment: The network controller evaluates the current channel usage and, based on the availability and the priority of the request, assigns a channel to the requesting user.

Data Transmission: The user transmits their data using the assigned channel for the duration of the communication.

Channel Release: Once the communication is complete, the channel is released back to the pool of available resources and can be reassigned to another user who requests it.

Advantages of DAMA
Efficiency: By allocating resources on-demand, DAMA systems can serve more users with fewer channels compared to fixed assignment schemes, leading to higher overall network efficiency.

Flexibility: DAMA can adapt to varying traffic loads, making it well-suited for networks with fluctuating usage patterns.

Fairness: It can prioritize requests based on predefined criteria, ensuring critical communications are given the necessary resources.

Applications of DAMA
Satellite Communications: DAMA is widely used in satellite networks, where bandwidth is a precious resource. It's particularly useful for supporting mobile users or remote locations where communication demands can vary significantly.

Military Communications: The flexibility and efficiency of DAMA make it suitable for military communication systems, which require robust and adaptive communication capabilities.

Mobile Radio Systems: DAMA can be used in terrestrial mobile radio systems to manage the allocation of channels to mobile units efficiently.

Considerations
Complexity: The dynamic nature of DAMA systems adds complexity to the network design, requiring sophisticated controllers capable of managing resource allocation in real-time.

Latency: The process of requesting and assigning channels can introduce latency, which might be an issue for time-sensitive communications.

Overhead: The signaling required for channel requests and assignments can consume bandwidth, reducing the capacity available for user data.

There are UHF, SHF, and EHF military satellites.
UHF satellites are used for tactical military use, which includes ground forces as well as those of the Navy and Air Force.
A footprint of a satellite is the total ground area for which a LOS path exists.
A UHF SATCOM transceiver can link with any satellite that includes the transceiver in its footprint.
The use of multiple footprints and ground relay stations can extend the SATCOM range to nearly the entire world.
The Satellite Management Center (SMC) regulates and assigns the satellite resources to users.
Directional antennas must be used with UHF SATCOM transceivers.
Demand Assigned Multiple Access (DAMA) is a way to timeshare available satellite resources in an efficient way.

COMMUNICATION SECURITY
COMSEC uses cryptography or scrambling to make information unintelligible to people who do not have a need to know or who should not know.
The security level of a COMSEC system depends on the mathematical complexity of the algorithms and the number of variables in the key.
Protection of the key is vital to securing the transmitted information.
Public key cryptography is widely used in the commercial sector.
Over-The-Air-Rekeying (OTAR) eliminates the need for manual loading of keys and provides a more secure method of key management.
TRANSEC protects the transmitted signal itself, to prevent signal detection or jamming of the transmission path.

Low Probability of Detection (LPD) systems use spread-spectrum and other techniques to “hide” the signal beneath the natural noise level.
Frequency-hopping radio systems jump rapidly in unison, from one frequency to another in apparently random patterns, using a common timing reference.
Presidio, CITADEL, and Sierra are modern COMSEC and TRANSEC engines

Digital Battlefield
Mobile communications vehicles provide hubs that receive and retransmit radio signals that cover the entire available spectrum. VHF Combat Net Radios (CNR) provide ground LOS communications between squads, while UHF radios provide the same for ground-to-air (G/A) close support.